Raghavan 2021 Pancreatic Ductal Adenocarcinoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → **GO**

20/20 PDAC samples (≥120 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 50 cells / 1 speck removed; iLISI median 1.00→7.22 of 20); both lobes (Ductal-body 6,494 / Cycling 674) carry the trajectory. One PANET sample (662 cells, a distinct neuroendocrine malignancy) was excluded to match the declared PDAC scope.

Gate 2 — module geography → **GO** (non-branching)

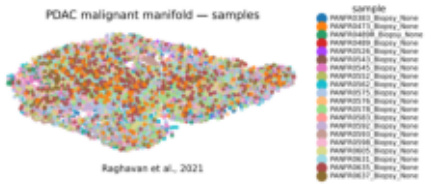
Single dominant pSom→Crest axis: pSom high over the classical-ductal body declines along pseudotime (ρ=−0.74); Crest anti-correlated (r=−0.68), depleted at the root and rising to a distal plateau (ρ=+0.61; crest-sm peaks mid-to-distal, decile pt=0.59). Gast co-varies with pSom (joint-high enrichment 1.15); joint pSom&Crest-high depleted (0.23) — an inverse gradient, not a hub.

Gate 3 — hub mutual exclusivity → **NA**

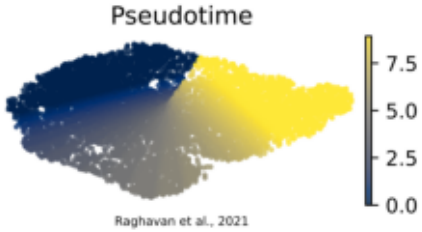
Single monotone axis — no bifurcation and no Mixed-type hub (curve: 2 tips, 0 forks, self-approach 2.57), so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

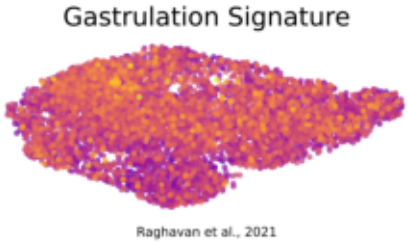
Gate 1: integration



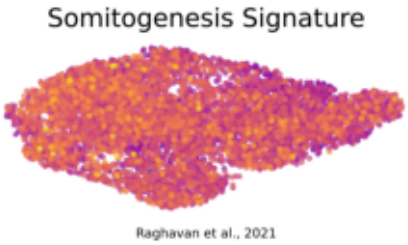
Trajectory / root



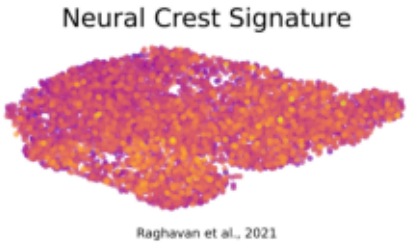
Gast



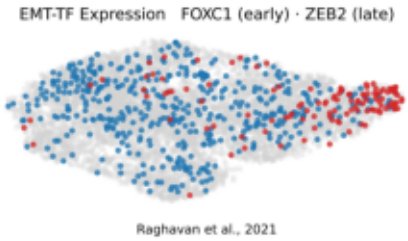
pSom



Crest



EMT-TFs



Dataset. PDAC (Raghavan et al., 2021), Seq-Well — 24 patients / 24,412 cells (master scope); malignant only: 7,904 cells, of which 7,242 are PDAC (662 PANET neuroendocrine cells dropped). After the ≥120-cell floor, 7,168 cells across 20 samples enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Raghavan's metastatic PDAC malignant manifold is a single monotone somitogenesis → distal-neural-crest gradient (classical-ductal body → neural-crest-like distal pole); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both lobes and is orthogonal to cell cycle (module–cyc |ρ|≤0.20; pt–cyc ρ=+0.11), so the whole manifold is kept. Raw module means are pSom-dominant and geography is a coherent inverse pSom→Crest gradient (Crest depleted at root, rising to a distal plateau) — not the mutually-exclusive territories of the failed Peng 2019 PDAC run. EMT-TFs FOXC1 (early/root) → ZEB2 (late) on the widened FOXC/PRRX panel.

Parameter choices

Min cells/sample	≥73 (geom-mean 120→45)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	PDAC only; whole manifold	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = pSom-high, Crest-low	**Compartments**	Ductal-body 6,494 / Cycling 674